Bilinear and Quadratic Forms: Answer Key

Problem sheet for the Day 10 class · CSIR NET previous year questions and exam-pattern problems

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	(D)	(A)	(A)	(A)	4	(D)	(A),(B),(C),(D)	(A),(C),(D)	(A),(B),(D)	(A),(B),(C)

Hints and short solutions

1. **(D)** CSIR NET Dec 2015 $Q = x^2 + y^2 + 2zw$ has eigenvalues $1, 1, 1, -1$: rank 4, signature $(3, 1)$. Forms equal to $Q(Pv)$ are exactly those of rank 4 and signature $(3, 1)$. $xy + z^2$ and $xy + y^2 + z^2$ omit w (rank ≤ 3); $x^2 + y^2 - zw$ has signature $(3, 1)$.
2. **(A)** Exam pattern The matrix is $J - I$ with J the all-ones matrix; J has eigenvalues $3, 0, 0$, so $J - I$ has $2, -1, -1$.
3. **(A)** Exam pattern Its matrix is $(1 - k)I + kJ$, with eigenvalues $1 + 2k$ (once) and $1 - k$ (twice). Both are positive iff $-\frac{1}{2} < k < 1$.
4. **(A)** Exam pattern By Sylvester's law of inertia a class is fixed by (p, n) with $p + n \le 3$: $1 + 2 + 3 + 4 = 10$ pairs.
5. **4** Exam pattern Its symmetric matrix is tridiagonal with zero diagonal and $\frac{1}{2}$ beside it; its determinant is $(\frac{1}{2})^4 = \frac{1}{16} \neq 0$. So the rank is 4.
6. **(D)** CSIR NET June 2018 $Q = x^2 + z^2 + 2xy - 2yz$ has symmetric matrix $S = \frac{1}{2}(A + A^t) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{pmatrix}$, with eigenvalues $2, 1, -1$. The Hessian is $2S \neq 2A$, the rank is 3, the signature is $(+\ +\ -)$, and an indefinite form has non-zero isotropic vectors.
7. **(A),(B),(C),(D)** CSIR NET Dec 2014 Symmetry gives $T_1(\alpha)T_2(\beta) = T_1(\beta)T_2(\alpha)$, so $T_2 = cT_1$ with $c \neq 0$ and $f = cT_1 \otimes T_1$. Then the rank is 1, the radical and the isotropic set are both $\ker T_1$ (dimension 2), and $f(\alpha, \alpha) = cT_1(\alpha)^2$ has the sign of c.
8. **(A),(C),(D)** CSIR NET June 2019 $\det(P^tAP) = -(\det P)^2 < 0$ rules out (B). Explicit P: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ gives $\mathrm{diag}(2, -2)$; $\begin{pmatrix} 1 & 1 \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$ gives $\mathrm{diag}(1, -1)$; $\begin{pmatrix} 1 & 1 \\ \frac{3}{2} & \frac{5}{2} \end{pmatrix}$ gives $\begin{pmatrix} 3 & 4 \\ 4 & 5 \end{pmatrix}$. (Booklet B; answer checked by direct computation.)
9. **(A),(B),(D)** CSIR NET June 2019 $a_{ij} = \int_0^1 x^{n_i} x^{n_j}\, dx$, so $v^tAv = \int_0^1 \left(\sum v_i x^{n_i}\right)^2 dx \ge 0$, with equality only for $v = 0$ exactly when the n_i are distinct. Equal n_i give a rank-1 matrix.
10. **(A),(B),(C)** Exam pattern (A) $B(e_i, e_j) = a_{ij}$. (B) $u^tAu = 0\ \forall u$ iff $A + A^t = 0$. (C) $A = \frac{A + A^t}{2} + \frac{A - A^t}{2}$, uniquely. (D) False: $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is non-degenerate and alternating.